

Therefore, answer is C.

17 Ans: A

Using Malus's Law, $I = I_0 \cos^2 \theta$,

Initially when polariser is vertical:

$$I_1 = I_0 \cos^2 0^\circ$$

After polariser is rotated by θ :

$$I_2 = I_0 \cos^2 \theta$$

Hence,
and

$$I_2 / I_1 = \cos^2 \theta$$

$$I_2 / I_1 = V_2 / V_1 = 2.20 / 3.50 = 0.6286$$

Therefore,

$$\cos \theta = \sqrt{0.6286} = 0.7928$$

$$\theta = 37.5^\circ$$

18 Ans: D